Is This the Real Life? Is This Just Fantasy? A Domain Adaptation Framework for Identifying Higher-Order Reality Representations

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Abstract

How does your brain know what's real? Neural responses to perceived versus imagined content often seem to overlap in their neuroanatomical loci and representational structure, but the brain is nevertheless able to differentiate between imagery and perceptual input. How does the brain engage in this reality monitoring capacity, and what can this tell us about the mechanisms behind the conscious experience of seeing? A class of theories of consciousness—Higher-Order (HO) theories—posits that conscious experience depends on the formation of higher-order representations, which monitor and interpret first-order (perceptual and imagined) representations. Yet HO theories differ in their positioning on the nature of such HO representations. Here, we develop and validate a computational approach to interrogating the nature of HO representations by using domain adaptation (DA), a machine learning technique that can statistically align data distributions from different sources. We apply DA to ask when and where patterns of human fMRI BOLD responses can be aligned across perception and imagination, and across contents, to reveal areas that encode content-agnostic reality signals, reality-agnostic content signals, and signals of one type that are conditioned on the other, thereby providing differential support for specific aspects of competing HO theories. We find content-conditioned reality signals in the precuneus and posterior cingulate cortex, while content-agnostic reality signals appear more in the occipital fusiform and occipital pole. In contrast, content representations specific to perception versus imagination appeared in the precuneus and intracalcarine cortex, while the temporal occipital fusiform cortex appeared to encode content regardless of its source. These results suggest highly localized and separable computations may collectively drive the overall assessment of whether an internal representation is real or imagined. Further, these results demonstrate the viability, promise, and power of DA approaches to interrogating the inputs to HO neural representations.

Introduction

Higher-Order (HO) theories propose that distinguishing perception from imagination requires HO representations beyond first-order sensory signals (Brown et al., 2019). However, they differ in how this distinction is encoded. Perceptual Reality Monitoring (PRM) theories propose a dedicated reality-monitoring signal separating perception from imagination (Lau, 2019, 2022), whereas Higher-Order State Space (HOSS) proposes that reality judgments emerge indirectly from representational strength or vividness without a separable reality dimension (Dijkstra & Fleming, 2023b; Fleming, 2020). Other accounts, such as HO Representation of a Representation (HOROR), suggest that HO representations may also re-represent first-order content, making reality monitoring inseparable from representational content (Brown, 2015). Distinguishing among these requires isolating neural representations related to reality monitoring, content, and vividness.

Here, we present a methodological proof of concept using Domain Adaptation (DA), a machine-learning approach that aligns data distributions across conditions (Orouji et al., 2024). We developed DA models that suppress differences either between perception and imagination or across stimulus contents, and combined them with classifiers applied to the non-aligned dimension. This framework tests whether content- and reality-related information remains decodable after DA, providing a way to assess whether these signals are separable or entangled.

Methods

Dataset. We used a public whole-brain fMRI dataset in which 18 participants viewed or imagined object categories (dogs vs. scissors) (Hebart et al., 2019). Analyses were performed in native space, with maps normalized to MNI space for visualization (Evans et al., 1993).



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Decoding. We performed two binary decoding analyses using logistic regression applied to multivoxel patterns: *content decoding* (dog vs. scissors across Perception/Imagery domains) and *reality decoding* (Perception vs. Imagery across content domains). We evaluated each within- and cross-domain with and without domain adaptation (DA) using balanced accuracy.

DA method and Searchlight. We applied the Subspace Alignment (SA) DA method (Fernando et al., 2014), which aligns principal component subspaces across domains. Whole-brain searchlight analyses used 12mm spheres, yielding voxel-wise maps of decoding performance and domain alignment.

DA Metrics. In addition to decoding accuracy, we quantified cross-domain mismatch using covariance distance (CovDist): the mean absolute difference between source and target covariance matrices within each searchlight sphere. To characterize DA effects, we computed normalized before:after ratios for CovDist and decoding accuracy (DcdRatio). We also computed Cohen's *d* effect sizes for decoding accuracy (d_{acc}) and covariance distance (d_{dist}) across train/test folds.

Interpretation. We defined criteria-based *interpretation categories* using baseline cross-domain decoding and DA-induced changes in decoding and CovDist in each searchlight. Briefly, regions with (1) *different but alignable* representations show weak baseline cross-domain decoding that improves after DA, indicating reduced CovDist and increased representational transferability; regions with (2) *similar (and alignable) representations* show above-threshold baseline cross-domain decoding that further improves after DA, suggesting at least partially shared representations across domains. Brain masks for each interpretation category were generated by applying pre-defined thresholds (based on prior analyses of the same dataset (Olza et al., 2025)) to decoding accuracy and distance metrics.

Results and Discussion

Group-level results were computed as voxel-wise overlap maps across subjects, excluding voxels present in fewer than four subjects. For content decoding, *different but alignable* representations (Fig. 1a) showed highest overlap in precuneus and intracalcarine cortex, suggesting content representations that differ across perception and imagination but become transferable after DA. In contrast, *similar (and alignable)* representations (Fig. 1b) appeared in temporal occipital fusiform cortex, consistent with partially shared or source-invariant content representations across perception and imagery.

For reality decoding, *different but alignable* representations (Fig. 1c) appeared mainly in precuneus, posterior cingulate cortex, and lateral occipital cortex, suggesting content-dependent reality representations that become transferable after alignment. In contrast, *similar (and alignable)* reality representations (Fig. 1d) were concentrated in occipital fusiform cortex, consistent with more content-invariant representations of perceptual source (reality vs. imagination).

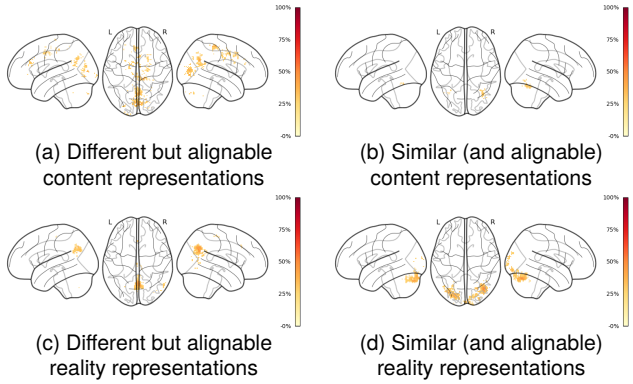


Figure 1: Group overlap maps for DA-based interpretation categories (color indicates % subjects per voxel).

Our DA framework testing when neural representations can be aligned across source of experience (perception vs. imagination) and stimulus content (dogs vs. scissors) was thus able to reveal candidate content-conditioned and content-agnostic reality signals, as well as reality-conditioned and reality-agnostic content signals. All four types were observed, in largely non-overlapping regions. These findings have implications for seeking any "downstream" system that performs reality monitoring: previous work suggests a prominent role for medial and prefrontal cortex in reality monitoring (Dijkstra et al., 2024; Dijkstra & Fleming, 2023a, 2023b), regions also implicated in HO representation formation. Our results refine this framework by isolating and characterizing candidate *inputs* to this reality-monitoring system or HO process, suggesting distributed and at least partially dissociable reality-monitoring representations rather than a single unified "reality signal."

Finally, improved decoding after DA suggests transferable representations across domains, whereas failure after alignment would have suggested more domain-specific coding. Here, we focused on positive changes in our metrics pre-to-post DA, to establish the viability of the DA approach. Future work can now use this technique to test the inseparability of HO dimensions, evidenced by *failure* to decode a target dimension after DA. The present findings are thus broadly compatible with HO theories proposing either separable or content-entangled reality representations. Moreover, because vividness ratings were unavailable, our findings cannot distinguish explicit reality coding from differences in overall representational strength, and therefore provide representational constraints rather than yet adjudicating among HO theories—all exciting topics for future.

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